Nano-Interface-Ready Neuro-Semantic State Communication for Privacy-Preserving Human-AI Cognitive Networks

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ABSTRACT

NeuroDigital Interfaces (NDI) provide a pathway for translating non-invasive neural observations into machine-interpretable cognitive states. Building on our published WITCOM 2025 NDAN chapter, this paper introduces a new neuro-semantic state communication layer for privacy-preserving human-AI cognitive networks. Instead of transmitting raw neural signals, the protocol communicates a task-relevant state estimate with uncertainty, consent scope, and provenance. The framework is nano-interface-ready rather than nanotechnology-validated: flexible electrodes, nanosensors, and low-power edge devices are defined as future sensing substrates, while the current validation layer remains software-based. We formalize state messages, governance constraints, calibration metrics, and a human-in-the-loop validation roadmap. We do not claim direct telepathy, a fabricated nanodevice, or clinical efficacy.

KEYWORDS

neurodigital interfaces, BCI, cognitive networking, nanosensors, human-AI interaction, neuro-rights

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1 INTRODUCTION

Human-AI interaction is increasingly constrained by the bandwidth and ambiguity of conventional input channels. NeuroDigital Interfaces can complement these channels by estimating task-relevant cognitive states from non-invasive measurements such as EEG or fNIRS. However, the engineering challenge is not simply signal classification: a useful interface must quantify uncertainty, adapt to changing users and environments, communicate safely with distributed agents, and preserve autonomy.

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This paper introduces Neuro-Semantic State Communication (NSSC) as a protocol layer beyond the prior NDAN publication. NSSC is not a claim of mind reading: it defines what a cognitive interface may communicate, how uncertainty is represented, how consent is scoped, and how downstream actions are audited. The nanotechnology connection is defined as a future sensing and packaging layer involving nanosensors, flexible electrode interfaces, and low-power edge electronics. Those components require independent hardware and biomedical validation and are not presented as experimentally demonstrated here.

2 RELATIONSHIP TO PRIOR WORK AND NEW CONTRIBUTION

The prior WITCOM 2025 chapter introduced the NDAN concept and adaptive cognitive-networking direction. This paper does not repeat that chapter's title, abstract, or claims. The new contribution is NSSC: a typed cognitive-state message, uncertainty and consent fields as protocol requirements, provenance and revocation rules, privacy-preserving state abstraction, and a validation matrix connecting future nanosensor requirements to network-level metrics.

3 SYSTEM ARCHITECTURE

NDAN is organized into five layers:

(1) **Sensing:** neural, physiological, and contextual observations;
(2) **Signal processing:** filtering, artifact detection, feature extraction, and calibration;
(3) **Cognitive state estimation:** probabilistic estimates of attention, workload, fatigue, and task intent;
(4) **Adaptive networking:** state-aware communication among human, edge, and AI agents;
(5) **Governance:** consent, provenance, access control, uncertainty disclosure, and reversibility.

The resulting loop is:

$$y_t \to \phi(y_t) \to p(z_t|\phi) \to m_t \to a_t \to y_{t+1}, \tag{1}$$

where y_t is a sensor observation, z_t is a latent cognitive state, m_t is a privacy-preserving message, and a_t is an adaptive system action.

4 NANOTECHNOLOGY-READY INTERFACE

The architecture identifies three interfaces where nanoscale engineering may provide future benefits:

- flexible, high-density electrode or nanosensor arrays;
- low-noise analog front ends and energy-aware edge processing;

- biocompatible packaging and reliable skin/interface contact.

These are requirements, not claims of a fabricated device. A valid hardware study must report material composition, sensor calibration, impedance, safety limits, sampling characteristics, and repeatability across users.

5 NEURO-SEMANTIC STATE COMMUNICATION

Rather than transmitting raw neural signals, NDAN transmits a state estimate and uncertainty:

$$m_t = (\hat{z}_t, \Sigma_t, c_t, \pi_t), \tag{2}$$

where $\hat{z}_t$ is the estimated state, Σ_t its uncertainty, c_t a consent scope, and π_t a provenance record. An agent may act only when confidence and consent satisfy policy constraints. State messages are revocable and purpose-limited: a state collected for interaction cannot automatically be reused for surveillance or consequential decisions.

6 EVALUATION PROTOCOL

The proposed validation sequence contains four stages:

(1) synthetic signals for unit and integration testing;
(2) public EEG/BCI datasets with subject-wise splits;
(3) controlled laboratory sessions with informed consent;
(4) hardware-in-the-loop evaluation of the sensing and edge pipeline.

Metrics include balanced accuracy, calibration error, false activation rate, end-to-end latency, energy per inference, bandwidth per cognitive-state message, privacy leakage, and user-reported workload. No result should be generalized across users without subject-wise validation.

Table 1: Required reporting matrix for future validation.

Dimension	Required evidence
Signal	Sampling, filtering, artifacts, sensor calibration
Model	Architecture, training split, uncertainty method
Network	Message size, latency, loss, adaptation policy
Safety	Consent, withdrawal, privacy, failure-safe behavior
Hardware	Materials, impedance, power, thermal limits
Reproducibility	Seeds, code, dataset license, environment

7 GOVERNANCE AND NEURO-RIGHTS

NDAN must not infer sensitive attributes without explicit purpose and consent. The governance layer applies data minimization, local processing where possible, cryptographic provenance, user-visible uncertainty, and an emergency stop. A cognitive estimate is not a diagnosis, legal fact, or authorization to act. Human-in-the-loop review is mandatory for consequential actions.

8 DISCUSSION

NSSC connects nanoscale sensing, BCI signal processing, adaptive networks, and human-AI systems without collapsing them into an unsupported telepathy claim. The strongest new contribution is the protocol layer for uncertainty, consent, revocation, and provenance. The most important research risk is domain shift: a model that works on synthetic signals may fail across users, electrode placements, motion artifacts, or changing tasks.

9 CONCLUSION

This paper defines a nano-interface-ready protocol for Neuro-Semantic State Communication as a new evolution beyond the published NDAN work. It provides a disciplined path from software prototypes to sensor and human-in-the-loop validation. Future work will implement public-dataset benchmarks, integrate the FCSTN/NDAN software lineage after its BCI test failures are corrected, and evaluate privacy-preserving cognitive-state communication.

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